Course code	Course title	L	T	P	J	C
CSE1001	Problem Solving and Programming	3	0	2	0	4

Pre-requisite NIL Syllabus version 1.0

Course Objectives:

1. To understand the basic concepts of problem solving using algorithms and flowcharts.
2. To develop programs using control structures and functions in a high level language.
3. To apply arrays, strings and pointers to solve engineering problems.

Course Outcomes:

On completion of this course, the student will be able to:
1. Explain the steps in problem solving and represent solutions using algorithms and flowcharts.
2. Write programs using decision making statements, loops and functions.
3. Develop programs that process arrays and strings.
4. Use pointers and structures to build modular programs.

Module:1 **Introduction to Problem Solving** 6 hours

Problem solving strategies, algorithms, flowcharts, pseudocode, structure of a program, variables, data types, operators and expressions.

Module:2 **Control Structures and Functions** 9 hours

Decision making statements, loops, nested loops, break and continue, functions, parameter passing, recursion.

Module:3 **Arrays and Strings** 8 hours

One and two dimensional arrays, searching and sorting, string handling functions, array of strings.

Module:4 **Pointers and Structures** 8 hours

Pointers, pointer arithmetic, dynamic memory allocation, structures, unions, enumerations.

Module:5 **Files** 8 hours

File operations, sequential and random access files, command line arguments.

Module:6 **Contemporary Issues** 2 hours

Guest lecture by industry experts.

Total Lecture hours: 45 hours

Text Book(s)

1. Yashavant Kanetkar, Let Us C, 16th Edition, BPB Publications, 2019.
2. Brian W. Kernighan and Dennis M. Ritchie, The C Programming Language, 2nd Edition, Pearson, 2015.

Reference Books

1. Herbert Schildt, C: The Complete Reference, 4th Edition, McGraw Hill, 2017.
2. E. Balagurusamy, Programming in ANSI C, 8th Edition, McGraw Hill, 2019.

Mode of Evaluation: CAT, Assignment, Quiz, FAT

List of Challenging Experiments (Indicative)

1. Programs using input and output statements and arithmetic operators.
2. Programs on decision making statements.
3. Programs using loops: pattern printing, series generation.
4. Programs on one dimensional and two dimensional arrays.
5. String manipulation programs without using library functions.
6. Programs using pointers and dynamic memory allocation.

Total Laboratory Hours 30 hours

Recommended by Board of Studies 12-08-2022
Approved by Academic Council No. 66, Date 16-12-2022

MAT2001 Statistics for Engineers L T P J C 3 0 0 0 3

Pre-requisite: Calculus and Matrices

Course Educational Objectives

The course aims to:
• Provide the basic concepts of probability and random variables.
• Introduce sampling distributions and estimation.
• Teach hypothesis testing and regression analysis.

Course Learning Outcomes

Upon successful completion of the course, students will be able to:
CO1: Compute probabilities and expectations of random variables.
CO2: Estimate population parameters using sampling distributions.
CO3: Perform tests of hypotheses and fit regression models.

UNIT I: Probability and Random Variables (9 hours)
Axioms of probability, conditional probability, Bayes theorem, discrete and continuous random variables, expectation, variance, moment generating functions.

UNIT II: Sampling and Estimation (9 hours)
Sampling distributions, central limit theorem, point estimation, confidence intervals.

UNIT III: Testing of Hypotheses (9 hours)
Large sample tests, t-test, chi-square test, F-test, analysis of variance.

UNIT IV: Correlation and Regression (9 hours)
Scatter diagrams, correlation coefficient, simple linear regression, multiple regression, time series.

Total Hours: 36

Text Books

1. Ronald E. Walpole, Probability and Statistics for Engineers and Scientists, 9th Edition, Pearson, 2012.
2. Jay L. Devore, Probability and Statistics for Engineering and the Sciences, 9th Edition, Cengage, 2016.

Reference Books

1. Sheldon Ross, A First Course in Probability, 9th Edition, Pearson, 2014.